H2O.ai

H2O.ai University

# The AI Lifecycle with the H2O.ai Platform: From Data to Production

SUPPLEMENT SLIDE DECK

# WHAT YOU WILL LEARN

### Data & Features

- Automate data preparation, profiling, and synthetic data generation
- Manage scalable, versioned feature stores for enterprise ML

### Predictive ML

- Train and track models using automated machine learning (AutoML)
- Explain and audit model behavior with SHAP, K-LIME, and DIA

### Enterprise MLOps

- Deploy, monitor, and manage real-time and batch model rollouts
- Register models, track experiments, and detect production drift

### Generative AI & Agents

- Build RAG pipelines, evaluate LLMs, and orchestrate multi-agent workflows
- Fine-tune LLMs and secure them with enterprise guardrails

### Governance

- Enforce compliance, Responsible AI, and audit readiness at scale

# COURSE INTRODUCTION & PLATFORM OVERVIEW

Most enterprise AI programs stitch together disconnected tools – one for data prep, another for modeling, another for deployment. H2O.ai takes a different approach.

- **What Makes It Different**

- Single platform architecture spanning the entire AI lifecycle
- Same governance model and security controls across all components
- Supports both predictive ML and generative AI within one environment
- Designed from the ground up for enterprise-scale operation

# PLATFORM COMPONENTS AT A GLANCE

### ▸ Predictive AI

•H2O Driverless AI — AutoML, feature engineering, explainability, AutoDoc
•H2O Feature Store — Feature registration, versioning, online/offline serving
•H2O MLOps — Model registry, deployment, experiment tracking, drift monitoring

### ▸ Generative AI

•Enterprise h2oGPTe — RAG, prompt engineering, guardrails, agents
•H2O Enterprise LLM Studio — LLM fine-tuning and instruction tuning
•H2O Eval Studio — LLM and RAG evaluation with LLM-as-a-judge

### ▸ Infrastructure

•H2O AI Engine Manager — Kubernetes-native compute orchestration
•H2O Orchestrator — Workflow scheduling and batch execution

H2O DRIVERLESSAI

# AUTOMATED DATA PREP & TRACEABILITY

H2O.ai

- H2O Driverless AI automates data wrangling directly within the ML training pipeline.
- Automated data profiling identifies data quality issues before modeling begins.
- The platform automatically captures data lineage, handles sensitive data, and enforces access controls.
- Preprocessing logic is embedded into scoring pipelines to prevent train-serve skew.

H2O FEATURE STORE

# FEATURE STORE & ENGINEERING

- The H2O Feature Store manages, versions, and serves ML features at enterprise scale.
- Features are transformed once, versioned, and synchronized between offline and online environments.
- Driverless AI evaluates hundreds of candidates to automate feature engineering.
- High-performing features are promoted to a searchable catalog for team discoverability.

H2O DRIVERLESS AI

# AUTOML, EXPLAINABILITY & BIAS TESTING

H2O.ai

- Driverless AI automatically selects algorithms, hyperparameters, and feature engineering strategies.
- The platform automatically generates explainability visualizations, including SHAP and K-LIME.
- Disparate Impact Analysis (DIA) tests model fairness across different demographic groups.
- AutoDoc generates automated model documentation for regulated industries.

H2O MLOps

# ENTERPRISE MLOPS DEPLOYMENT

- Models are registered directly into H2O MLOps from Driverless AI.
- Deployment options include real-time REST endpoints, batch scoring, and external deployments.
- A/B testing and Champion/Challenger configurations ensure safe model rollouts.
- Containerized deployments and autoscaling seamlessly handle variable workloads.

H2O DRIVERLESS AI

H2O MLOps

# REGISTRY, ARTIFACTS & EXPERIMENT TRACKING

- The enterprise registry stores scoring artifacts, training metrics, and custom metadata.
- Third-party MLflow models can be imported and managed alongside native H2O models.
- Every model is linked directly to its originating experiment, dataset, and artifacts.
- Parallel experiments are tracked, compared, and synced for complete reproducibility.

H2O MLOps

# DRIFT DETECTION & MONITORING

- Model monitoring is critical to maintaining production accuracy over time.
- Users can easily configure baseline metrics directly in H2O MLOps.
- Statistical aggregates are used to calculate data and feature drift detection.
- Apache Superset enables teams to build custom, granular monitoring dashboards.

Enterprise h2oGPTe

# PROMPT ENGINEERING & LLM FINE-TUNING

H2O.ai

- Enterprise h2oGPTe provides a centralized prompt library to clone and share templates.
- Multilingual templates ensure consistent AI behavior across global markets.
- Fine-tuning is utilized for specialized, domain-specific tasks where frontier models underperform.
- H2O Enterprise LLM Studio manages the instruction tuning workflow and model export.

Enterprise h2oGPTe

# LLM GUARDRAILS & SAFETY

H2O.ai

- A multi-stage guardrail architecture evaluates content at ingestion, prompt, and response.
- A defense-in-depth approach handles PII detection and redaction using Regex, Presidio, and ModernBERT.
- PromptGuard actively detects and blocks adversarial jailbreak attempts.
- All guardrail violations are logged to support compliance audits.

H2O EVAL STUDIO

# RAG & LLM EVALUATION

H2O.ai

- RAG (Retrieval Augemented Generation) is necessary to ground LLMs in enterprise knowledge and prevent hallucinations.
- Hybrid search maximizes recall using semantic similarity, BM25, and cross-encoder reranking.
- Multimodal RAG allows users to query across text, images, audio, and video.
- H2O Eval Studio provides systematic evaluation frameworks, including LLM-as-a-judge.

Enterprise h2oGPTe

Enterprise h2oGPTe

# ORCHESTRATION, APIS & SDKS

H2O.ai

- H2O.ai natively orchestrates all platform workloads on Kubernetes for automated scaling.
- Administrators define resource profiles, keeping infrastructure spend controlled.
- Python SDKs and REST APIs are available to extend the platform programmatically.
- The Model Context Protocol (MCP) enables agents to securely invoke external tools and services.

# AI AGENTS & AGENTIC WORKFLOWS

- The H2O Data Science Agent autonomously performs EDA, configures models, and analyzes SHAP values.
- The Agent Builder translates natural language workflows into production-ready Python code.
- The A2A protocol standardizes communication for multi-agent ecosystems.
- The H2O Super Agent acts as a natural language IDE to execute tasks and write code.

# VISUAL PIPELINES & WORKFLOW COMPOSITION

- H2O.ai supports visual, no-code, and code-first development styles without locking teams in.
- Driverless AI utilizes visual pipeline diagrams to map out complex feature engineering steps.
- UI-based, no-code workflows can be exported as fully executable Python code.
- h2oGPTe agents generate and export sandbox-tested Python code during execution.

H2O DRIVERLESSAI

# GOVERNANCE, AUDIT & BUSINESS ROI

H2O.ai

- RBAC and workspaces enforce governance directly at the platform architecture level.
- Model constraints embed regulatory logic directly into model behavior during training.
- Centralized audit logs capture every user action and operational event for regulatory examinations.
- Custom scoring functions optimize models for actual business value instead of standard statistical metrics.

Enterprise h2oGPTe

H2O SUPER AGENT™

# HELPFUL RESOURCES

- **H2O.ai Platform Overview**:
  https://h2o.ai/

- **H2O Driverless AI (AutoML)**:
  https://h2o.ai/platform/ai-cloud/make/h2o-driverless-ai/

- **Enterprise h2oGPTe (GenAI & Agents)**:
  https://h2o.ai/platform/enterprise-h2ogpte/

- **H2O Eval Studio (LLM Safety & Evaluation)**:
  https://h2o.ai/platform/enterprise-h2ogpte/eval-studio/

- **H2O Documentation Hub**:
  https://docs.h2o.ai/

H2O.ai

H2O.ai University

# The End

Thank You

THE AI LIFECYCLE WITH THE H2O.AI PLATFORM:
FROM DATA TO PRODUCTION